

Synthèse : rangement de modèles

Consommation d'énergie

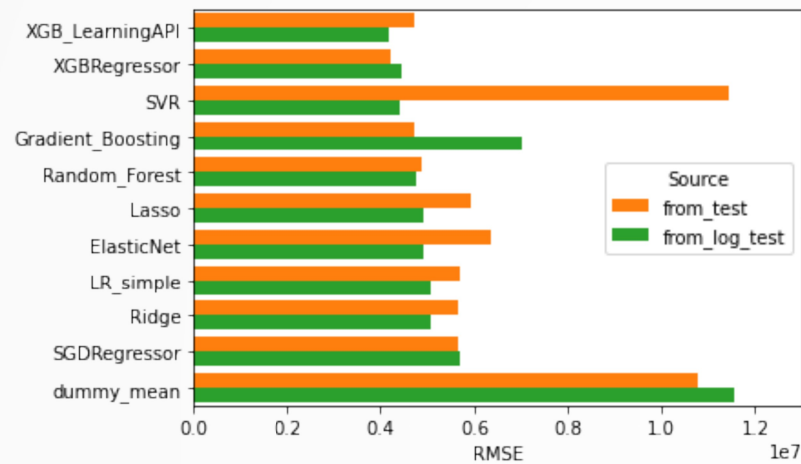
	Model	Source	Variables	RMSE	MAE	R2Score	Fit time
27	Random_Forest	from_log_test	with_score	2439986.714	1293687.064	0.913	4.270
32	SVR	from_log_test	with_score	2518081.310	1409721.345	0.907	0.012
29	XGB_LearningAPI	from_log_test	with_score	2619733.162	1352939.507	0.900	0.012
31	SGDRegressor	from_log_test	with_score	2817795.605	1503585.765	0.884	0.012
10	Random_Forest	from_test	with_score	2903857.745	1506811.064	0.877	2.389

Émissions CO₂

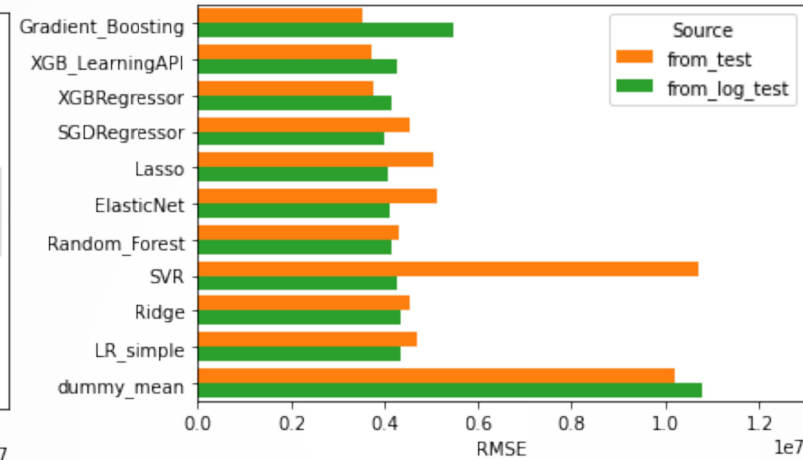
	Model	Source	Variables	RMSE	MAE	R2Score	Fit time
62	XGB_LearningAPI	from_log_test	without_score	121.534	53.150	0.713	0.010
45	Gradient_Boosting	from_test	without_score	124.860	54.755	0.697	2.783
41	ElasticNet	from_test	without_score	125.520	64.892	0.693	0.003
39	Lasso	from_test	without_score	126.667	65.562	0.688	0.002
63	XGBRegressor	from_log_test	without_score	128.259	55.583	0.680	0.010

- Le rangement n'est pas le même à chaque lancement du notebook: l'erreur finale dépend :
 - Fortement, du random state pendant la séparation initiale du jeu de test
 - Légèrement, de la séparation de données pendant validation croisée
 - Légèrement, de l'aléatoire dans les modèles aléatoires comme Random Forest et Gradient Boosting

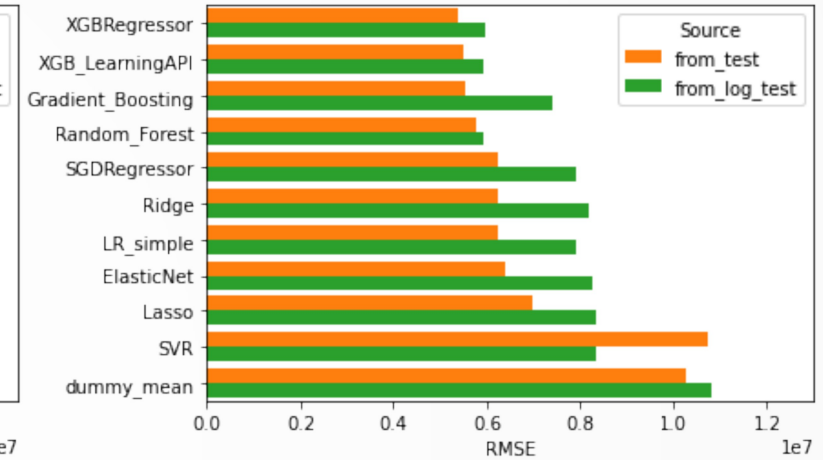
Énergie : l'utilité du passage de la cible au logarithme, avec Energy Star Score



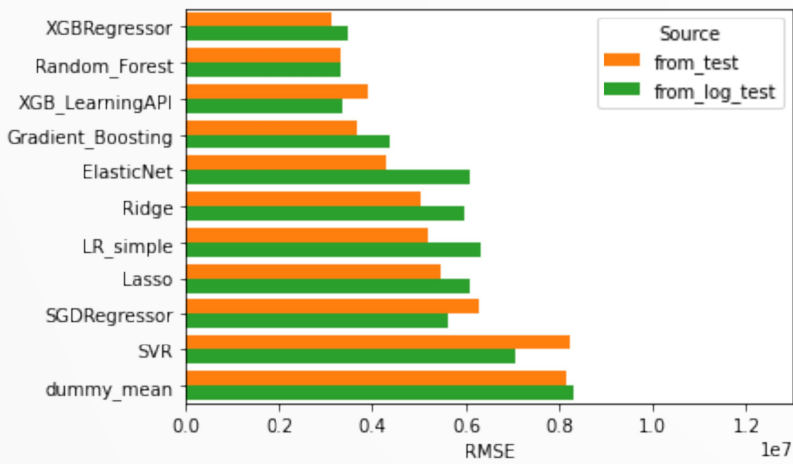
Random state = 1



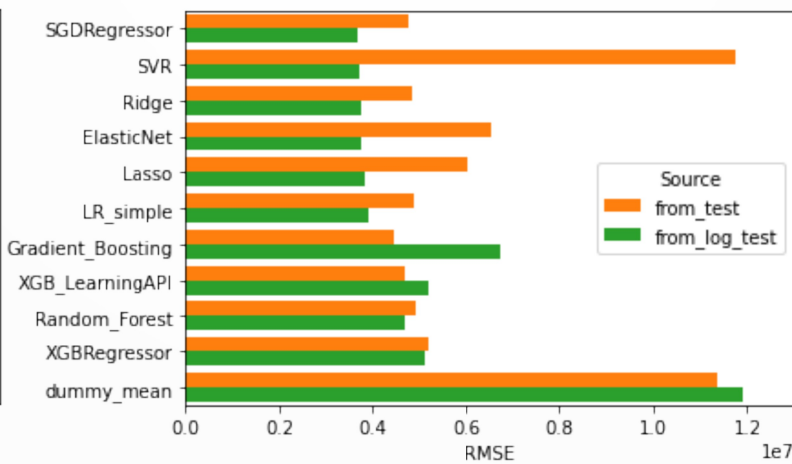
Random state = 2



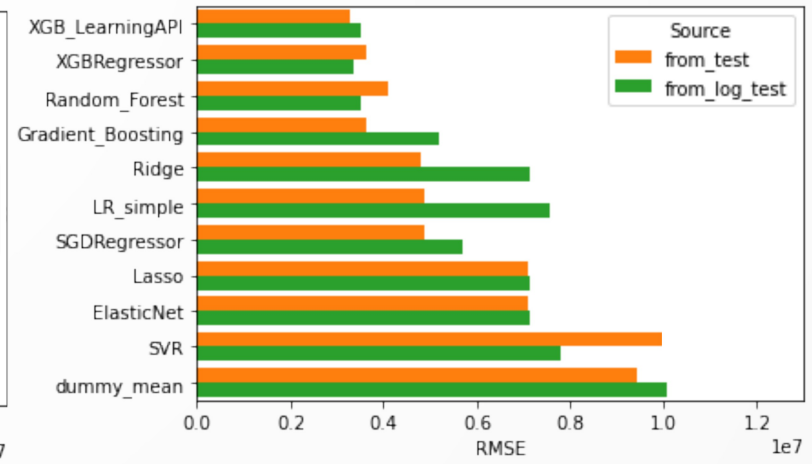
Random state = 3



Random state = 4

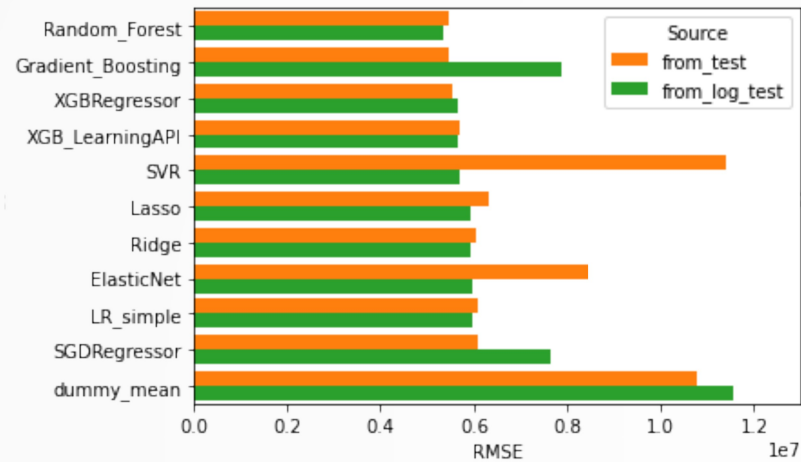


Random state = 5

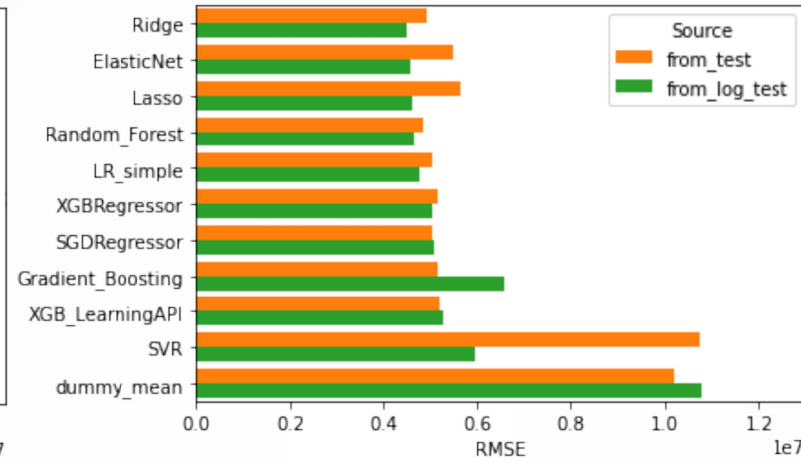


Random state = 6

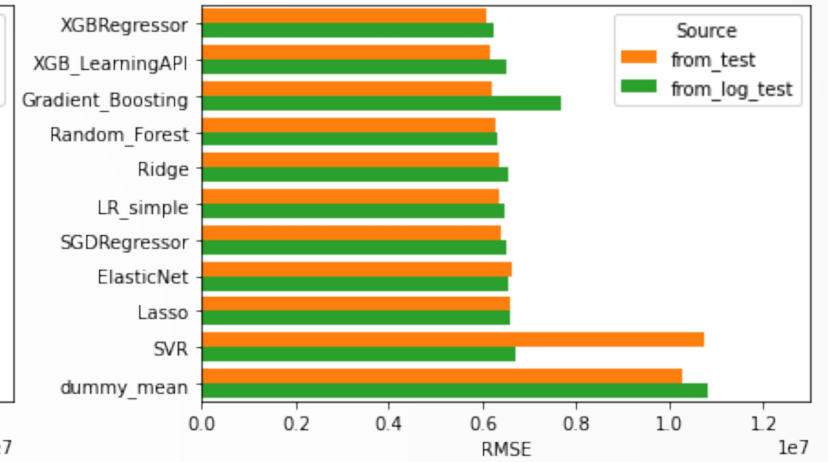
Énergie : l'utilité du passage de la cible au logarithme, sans Energy Star Score



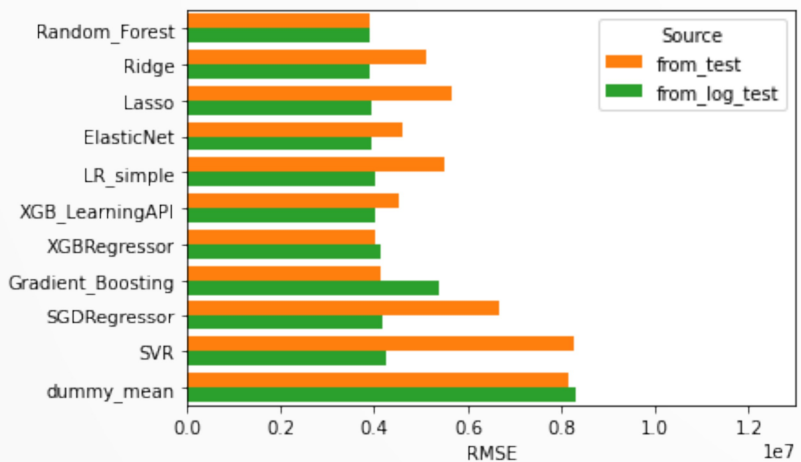
Random state = 1



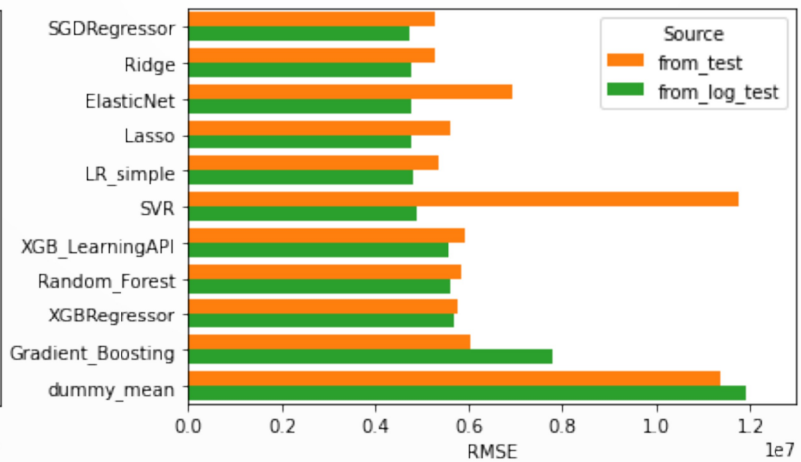
Random state = 2



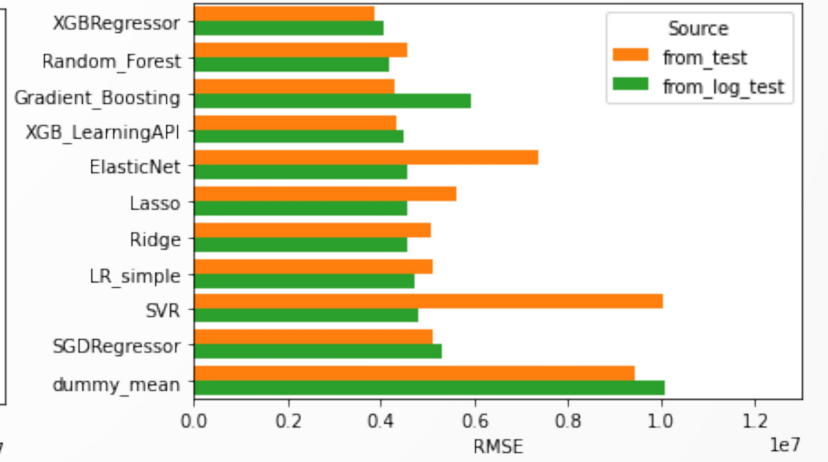
Random state = 3



Random state = 4



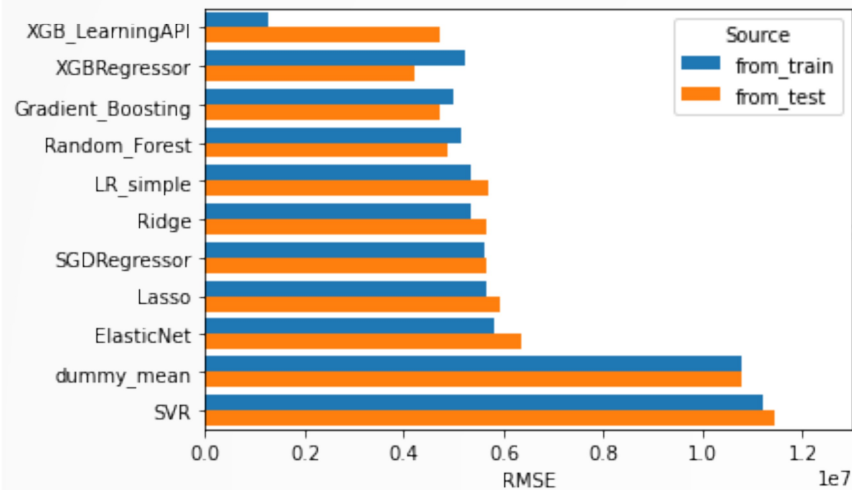
Random state = 5



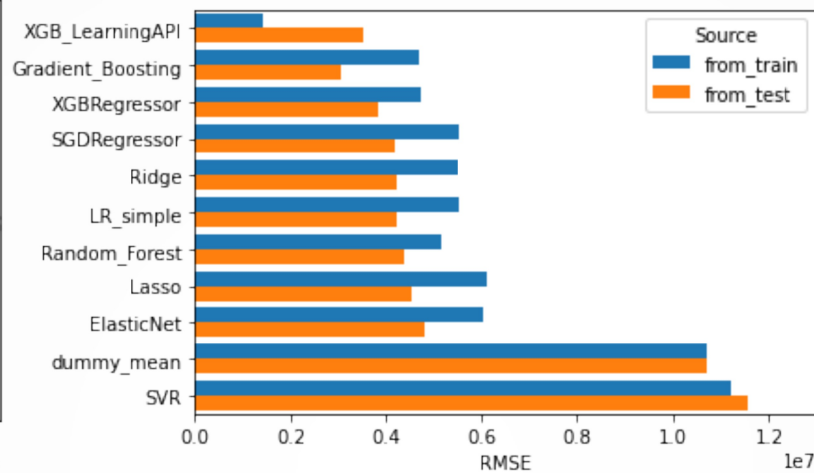
Random state = 6

Énergie : sur-apprentissage des modèles

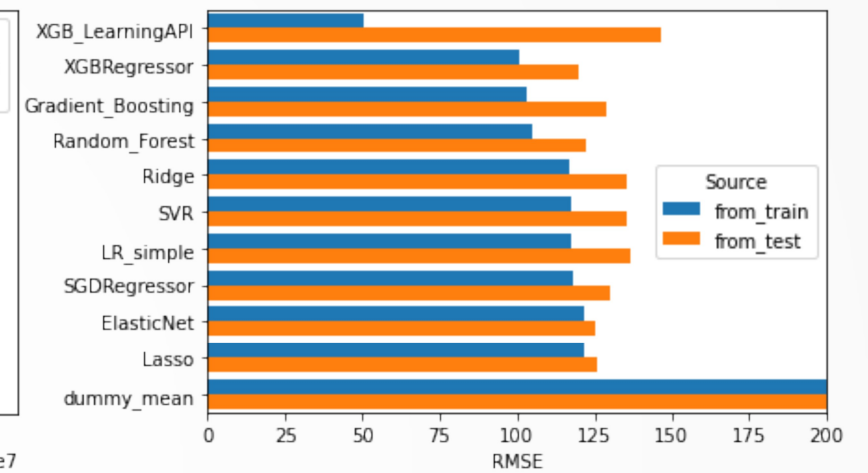
Avec Energy Star Score



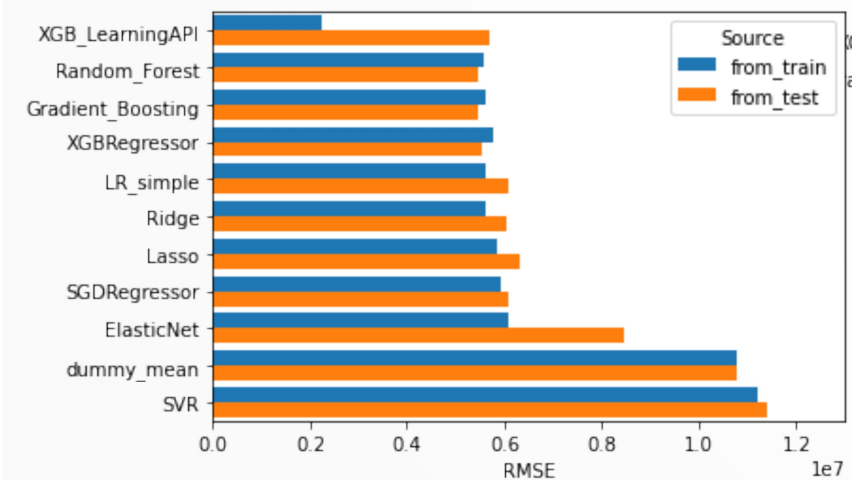
Avec Energy Star Score



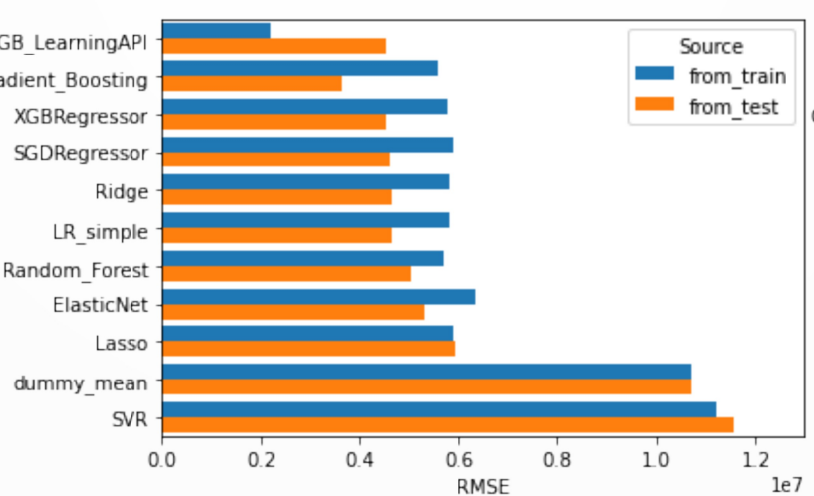
Avec Energy Star Score



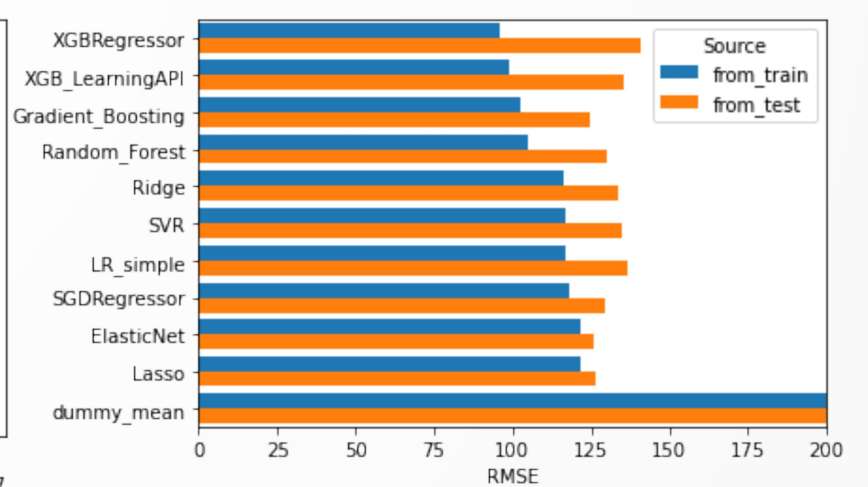
Random state = 1



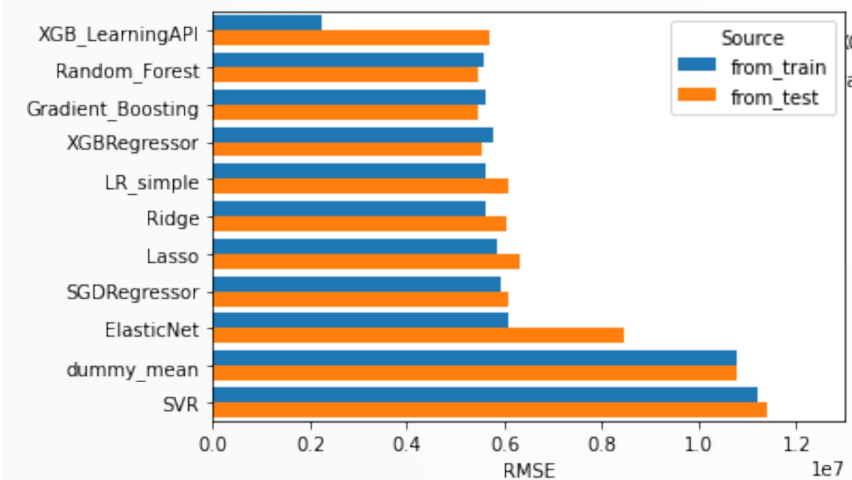
Random state = 2



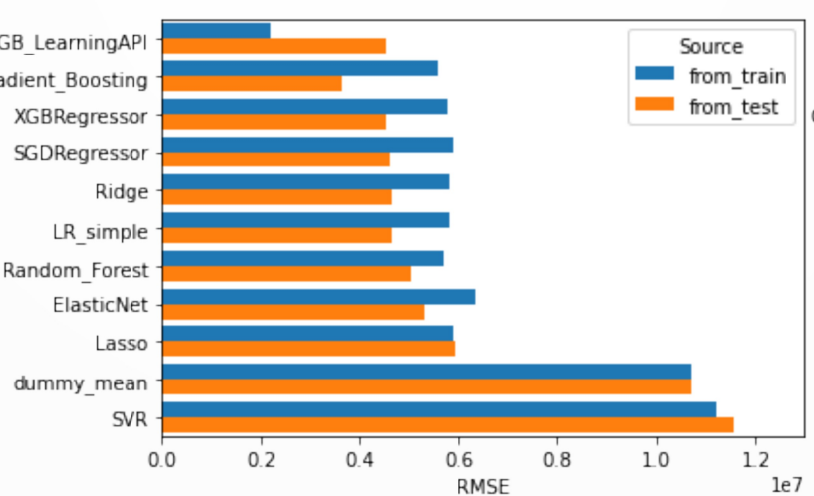
Random state = 3



Sans Energy Star Score



Sans Energy Star Score



Sans Energy Star Score

